ATAR · YEAR 12 CHEMISTRY · UNIT 3 · AREA OF STUDY 3

Redox, Galvanic Cells & Electrolysis

Study Notes aligned to Chapters 8, 9 & 10

Chapter 8 Redox reactions (8.1–8.4), Chapter 9 Galvanic cells (9.1–9.5) & Chapter 10 Electrolysis (10.1–10.2): oxidation & reduction, oxidation numbers, half-equations, the reactivity series, galvanic cells, the electrochemical series, predicting reactions, everyday power sources, corrosion, electrolytic cells, competition at electrodes, quantitative electrolysis & industrial applications.

CHAPTER 8 — REDOX REACTIONS

8.1 Oxidation and Reduction

A **redox reaction** transfers electrons between species. **Oxidation** and **reduction** always occur together — electrons lost by one species are gained by another.

OIL RIG

Oxidation **I**s **L**oss · **R**eduction **I**s **G**ain (of electrons)

THREE WAYS TO DESCRIBE REDOX (HISTORICAL → MODERN)

Process	Electrons	Oxygen	Hydrogen	Ox. number
Oxidation	lost	gained	lost	increases
Reduction	gained	lost	gained	decreases

Early definitions were based on gain/loss of **oxygen** (combustion, metal extraction); the modern, general definition is **electron transfer**.

Other examples of redox: combustion, corrosion, metal displacement, cellular respiration and photosynthesis, and metals reacting with acids are all electron-transfer reactions.

Oxidising agents and reducing agents

DEFINITIONS

- **Oxidising agent (oxidant):** accepts electrons → is itself **reduced**.
- **Reducing agent (reductant):** donates electrons → is itself **oxidised**.

COMMON CONFUSION

The oxidising agent is the species that gets **reduced**; the reducing agent is the species that gets **oxidised**.

Writing simple half-equations

A **half-equation** shows one process with its electrons. For simple metal/ion systems, add electrons to balance the charge:

Oxidation: $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$ (electrons on the right)

Reduction: $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$ (electrons on the left)

CHECK

Both atoms **and** charge must balance. The number of electrons equals the change in total charge across the half-equation.

8.2 Oxidation Numbers

The **oxidation number** is the charge an atom *would* have if all bonds were fully ionic — the tool for tracking electron transfer.

OXIDATION NUMBER RULES (PRIORITY ORDER)

1. Free element = **0** (Na, O_2, Cl_2, S_8, P_4).
2. Monatomic ion = its charge (Na^+=+1, S^{2-}=−2, Al^{3+}=+3).
3. Group 1 = +1; Group 2 = +2; Al = +3 in compounds.
4. F = −1 always. H = +1 (but −1 in metal hydrides, e.g. NaH).
5. O = −2 (but −1 in peroxides e.g. H_2O_2; +2 in OF_2).
6. Sum = **0** in a neutral compound; = **ion charge** in a polyatomic ion.

Calculating oxidation numbers

WORKED EXAMPLES

SO_4^{2-}: $x + 4(-2) = -2 \Longrightarrow \mathbf{S = +6}$

MnO_4^-: $x + 4(-2) = -1 \Longrightarrow \mathbf{Mn = +7}$

$Cr_2O_7^{2-}$: $2x + 7(-2) = -2 \Longrightarrow \mathbf{Cr = +6}$

NH_4^+: $x + 4(+1) = +1 \Longrightarrow \mathbf{N = -3}$

Using oxidation numbers to name chemicals (Stock names)

For metals with more than one common charge, the oxidation number is shown as a **Roman numeral** in the name:

Formula	Oxidation no.	Name
$FeCl_2$	Fe = +2	iron(II) chloride
$FeCl_3$	Fe = +3	iron(III) chloride
Cu_2O	Cu = +1	copper(I) oxide

CuO	Cu = +2	copper(II) oxide

Using oxidation numbers to identify oxidation & conjugate pairs

RULES

- Oxidation number **increases** → oxidised; **decreases** → reduced; **no change** → not redox.
- Each oxidant/reductant forms a **conjugate redox pair**: oxidised form + n e^- ⇌ reduced form (e.g. Cu^{2+}/Cu).

Disproportionation

One element is simultaneously oxidised and reduced:

$$Cl_2 + H_2O \rightarrow HOCl + HCl \quad (Cl: 0 \rightarrow +1 \text{ and } 0 \rightarrow -1)$$

8.3 More Complex Redox Equations

BALANCING A HALF-EQUATION IN ACIDIC SOLUTION — 5 STEPS

1. Balance atoms except O and H.
2. Balance **O** with **H_2O**.
3. Balance **H** with **H^+**.
4. Balance **charge** with **e^-**.
5. Check atoms and charge.

EXAMPLE — PERMANGANATE REDUCED IN ACID

$$\mathbf{MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O}$$

BASIC SOLUTION

Balance as if acidic, then add **OH^- to both sides** equal to the H^+; combine $H^+ + OH^- \rightarrow H_2O$ and cancel duplicate water.

Combining half-equations into a full equation

METHOD

Multiply each half-equation so the **electrons are equal**, add them, then cancel electrons and any identical species.

EXAMPLE — MNO_4^- OXIDISING FE^{2+} IN ACID

$$\text{Reduction } (\times 1)\text{: } MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$$

$$\text{Oxidation } (\times 5)\text{: } 5Fe^{2+} \rightarrow 5Fe^{3+} + 5e^-$$

$$\mathbf{Overall: MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O}$$

Charge check: left +17 = right +17 ✓

APPLICATION — THE DICHROMATE BREATHALYSER

Acidified orange dichromate oxidises ethanol in breath; chromium is reduced from +6 (orange $Cr_2O_7^{2-}$) to +3 (green Cr^{3+}). The orange→green colour change shows alcohol was present.

8.4 The Reactivity Series of Metals

Metals are **reducing agents** — they lose electrons to form positive ions. The more reactive a metal, the more readily it is oxidised, i.e. the **stronger a reducing agent** it is.

> **REACTIVITY (ACTIVITY) SERIES — MOST → LEAST REACTIVE**
>
> **K · Na · Ca · Mg · Al · Zn · Fe · (H) · Cu · Ag · Au**
>
> Metals above hydrogen displace H_2 from acids; the most reactive also react with water. This order is the reverse of the metals' positions in the electrochemical series (§9.2).

Metal displacement reactions

A **more reactive metal displaces the ion of a less reactive metal** from solution (the reactive metal is oxidised; the less-reactive ion is reduced):

$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s) \quad \text{(Zn displaces Cu)}$$

> **PREDICTING A DISPLACEMENT**
>
> Will iron displace silver? Fe is more reactive than Ag, so **yes**:
>
> $$Fe(s) + 2Ag^{+}(aq) \rightarrow Fe^{2+}(aq) + 2Ag(s)$$
>
> The reverse (Ag + Fe^{2+}) does **not** occur.

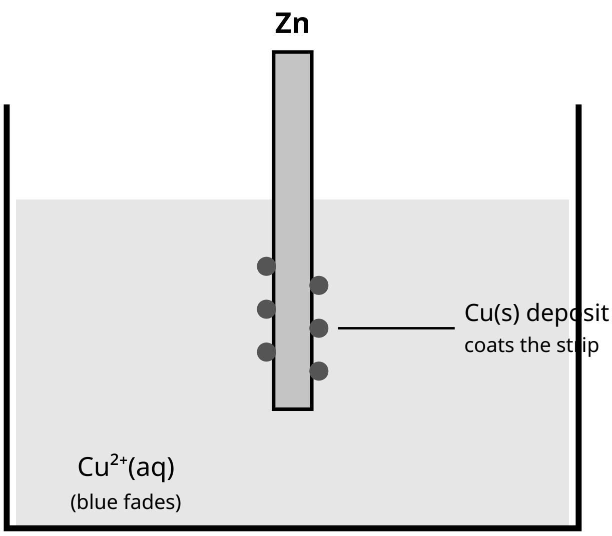

Figure 1. Metal displacement: a zinc strip in copper(II) sulfate. Zn (more reactive) is oxidised and dissolves ($Zn \rightarrow Zn^{2+} + 2e^-$); Cu^{2+} is reduced to copper metal that coats the strip ($Cu^{2+} + 2e^- \rightarrow Cu$). Overall $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$; the blue solution fades as Cu^{2+} is consumed.

Reactions of metals with water, acids and oxygen

Reactivity	With cold water	With dilute acid
Very reactive (K, Na, Ca)	react readily → metal hydroxide + H_2	violent → salt + H_2
Moderate (Mg, Al, Zn, Fe)	slow / only with steam	steady → salt + H_2
Unreactive (Cu, Ag, Au)	no reaction	no reaction (below H)

e.g. $2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$ · $Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

CHAPTER 9 — GALVANIC CELLS

9.1 Galvanic Cells

A **galvanic (voltaic) cell** converts chemical energy into electrical energy using a **spontaneous** redox reaction. Separating the two half-reactions forces electrons through an external wire — the current.

Half-cells

Each electrode + its solution is a **half-cell** (one oxidation, one reduction). If a half-cell has no solid conductor (e.g. Fe^{3+}/Fe^{2+}) an **inert electrode** (Pt or graphite) is used.

Electrode	Process	Galvanic polarity
Anode	Oxidation	negative (−)
Cathode	Reduction	positive (+)

MEMORY AIDS & MOVEMENT

- **AN OX** = ANode OXidation · **RED CAT** = REDuction CAThode.
- Electrons flow **anode → cathode** in the wire.
- In the salt bridge: **anions → anode**, **cations → cathode**.

Purpose of the salt bridge

The salt bridge (inert electrolyte, e.g. KNO_3) **completes the circuit** and **maintains electrical neutrality** in each half-cell by allowing ions to migrate. It does *not* carry electrons.

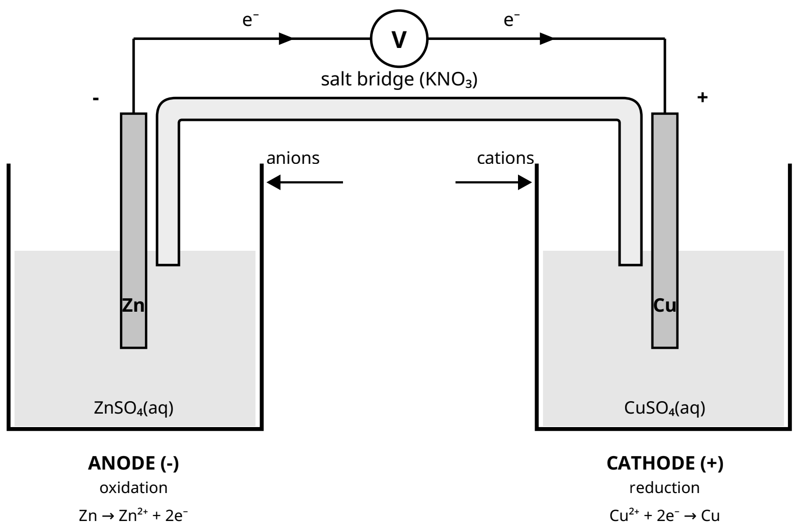

Figure 2. Galvanic (Daniell) cell. Electrons flow anode (−) → cathode (+); the salt bridge keeps each half-cell neutral (anions → anode, cations → cathode).

Writing an overall equation for a cell reaction

DANIELL CELL

Anode: $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

Cathode: $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

Overall: $\mathbf{Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)}$

Shorthand cell notation

Anode on the left. **|** = phase boundary; **‖** = salt bridge.

$$Zn(s) \mid Zn^{2+}(aq) \parallel Cu^{2+}(aq) \mid Cu(s)$$

9.2 The Electrochemical Series

Half-cell potentials are measured against the **Standard Hydrogen Electrode (SHE)**, defined as exactly **0.00 V**. The ordered list of standard reduction potentials is the **electrochemical series**.

STANDARD CONDITIONS (THE "°")

Solutions **1.0 mol L^{-1}** · gases **100 kPa** · **25 °C** · measured vs the SHE.

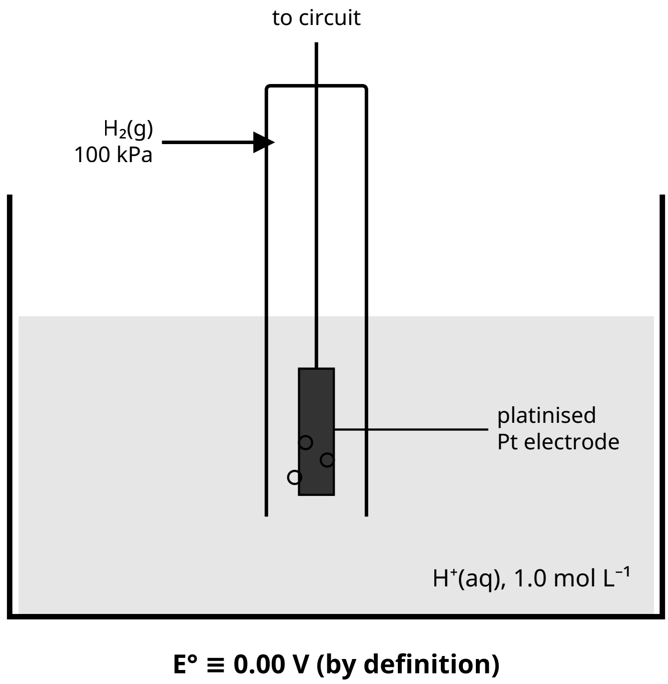

Figure 3. The Standard Hydrogen Electrode (SHE): H_2 at 100 kPa over inert Pt in 1.0 mol L^{-1} H^+ at 25 °C, assigned E° = 0.00 V so all other potentials can be measured against it.

Standard electrode potentials

READING THE SERIES (ALL VALUES ARE REDUCTIONS)

- **More positive E°** → stronger **oxidising agent** (top-left; e.g. F_2, MnO_4^-).
- **More negative E°** → its reduced form is a stronger **reducing agent** (bottom-right; e.g. Li, K, Na).

Predicting cell reactions & calculating the voltage

CELL EMF

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$$

Both values taken as reductions from the series; the half-cell with the higher E° is the cathode (reduced). A positive E°_{cell} confirms a spontaneous cell reaction.

EXAMPLE — ZN | CU CELL

$E^\circ_{cell} = E^\circ(Cu^{2+}/Cu) - E^\circ(Zn^{2+}/Zn) = (+0.34) - (-0.76) = \mathbf{+1.10\ V}$

E° DOES NOT SCALE WITH AMOUNT

E° is **intensive** — when you multiply a half-equation to balance electrons, do **not** multiply its E° value.

9.3 Predicting Direct Redox Reactions

The electrochemical series predicts whether two species react **directly** (mixed together, not in a cell).

THE RULE

A species reacts if the **oxidant (left side) has a higher E° than the reductant (right side)** it is mixed with — i.e. the higher-E° species is reduced and the lower-E° species is oxidised. Equivalent test: $\mathbf{E^\circ_{cell} = E^\circ_{oxidant} - E^\circ_{reductant} > 0} \Longrightarrow$ reaction occurs.

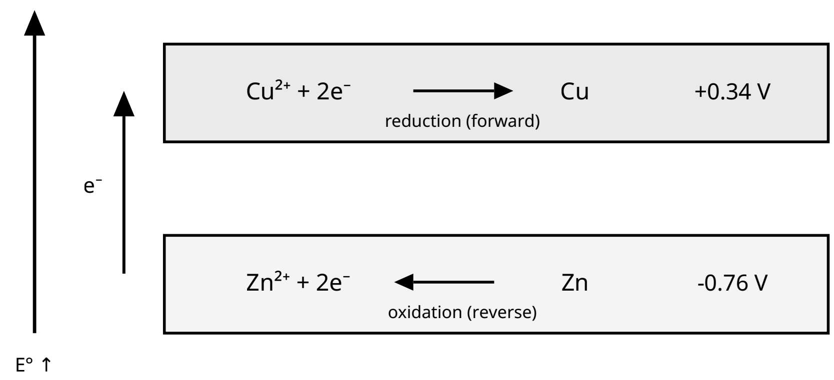

Figure 4. Predicting a reaction from the electrochemical series (the "anticlockwise rule"). The couple with the higher E° (top) undergoes reduction; the lower-E° couple (bottom) is reversed to oxidation. Electrons pass from the reductant (Zn) up to the oxidant (Cu^{2+}), so $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$ is spontaneous.

WORKED EXAMPLE — WILL CL_2 OXIDISE BR^-?

Cl_2/Cl^- (+1.36 V) is higher than Br_2/Br^- (+1.09 V), so Cl_2 is reduced and Br^- is oxidised:

$Cl_2(aq) + 2Br^-(aq) \rightarrow 2Cl^-(aq) + Br_2(aq) \quad E^\circ_{cell} = +0.27\ V$

(The reverse — $Br_2 + Cl^-$ — does not occur.)

Limitations of predictions

- Valid only at **standard conditions**; concentration, temperature, pressure shift the real potential.
- Predicts **feasibility, not rate** — a spontaneous reaction may be very slow.
- Values assume aqueous solutions; different solvents/complexes can change behaviour.

9.4 Everyday Sources of Power

Type	Feature	Examples
Primary	Non-rechargeable; reaction not reversible	dry (Leclanché) cell, alkaline cell
Secondary	Rechargeable; reaction reversed by an external supply	lead–acid accumulator, lithium-ion cell
Fuel cell	Reactants supplied continuously; not stored in the cell	hydrogen–oxygen fuel cell

ALKALINE CELL (PRIMARY)

Anode: $Zn(s) + 2OH^- \rightarrow ZnO(s) + H_2O + 2e^-$

Cathode: $2MnO_2(s) + H_2O + 2e^- \rightarrow Mn_2O_3(s) + 2OH^-$

LEAD–ACID ACCUMULATOR (SECONDARY)

Anode: $Pb(s) + SO_4^{2-} \rightarrow PbSO_4(s) + 2e^-$

Cathode: $PbO_2(s) + 4H^+ + SO_4^{2-} + 2e^- \rightarrow PbSO_4(s) + 2H_2O$

Recharging drives these reactions in reverse (this is electrolysis — Chapter 10).

LITHIUM-ION CELL (SECONDARY)

Rechargeable cell in phones, laptops and EVs. Li^+ ions shuttle between a graphite anode and a lithium-metal-oxide cathode (e.g. $LiCoO_2$) during discharge/charge:

Anode: $LiC_6 \rightarrow C_6 + Li^+ + e^-$ Cathode: $CoO_2 + Li^+ + e^- \rightarrow LiCoO_2$

High energy density and light weight; uses a lithium-salt organic (non-aqueous) electrolyte.

HYDROGEN–OXYGEN FUEL CELL (ACIDIC)

Anode: $2H_2 \rightarrow 4H^+ + 4e^-$ Cathode: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$

Overall: $2H_2 + O_2 \rightarrow 2H_2O$

Because chemical energy is converted **directly** to electricity (not via heat/combustion), fuel cells are highly efficient and produce only water. Challenges: producing H_2 cleanly and storing it safely (compressed, cryogenic, or in metal hydrides).

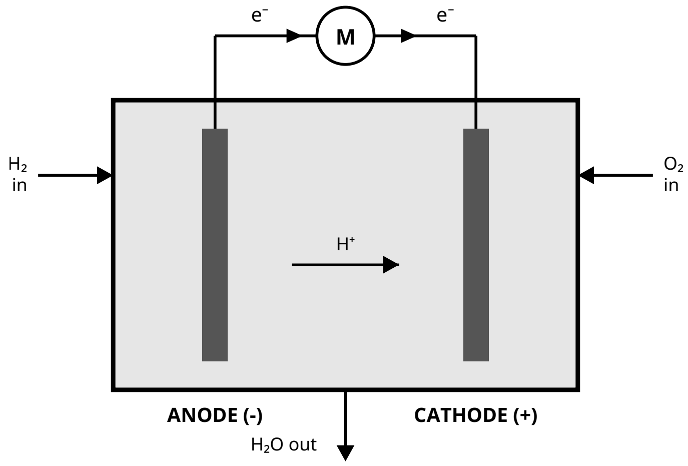

Figure 5. Hydrogen–oxygen fuel cell. Anode: $2H_2 \rightarrow 4H^+ + 4e^-$; cathode: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$. Electrons flow anode → cathode through the load (M); H^+ migrates through the electrolyte. Overall $2H_2 + O_2 \rightarrow 2H_2O$ — the only product is water.

9.5 Corrosion

Corrosion is the unwanted oxidation of a metal. **Wet corrosion** (rusting of iron) is electrochemical and needs **both water and oxygen**; it is faster with dissolved salts or acids.

WET CORROSION PROCESS

Anodic: $Fe(s) \rightarrow Fe^{2+}(aq) + 2e^-$

Cathodic: $O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$

Fe^{2+} is further oxidised by O_2 to hydrated iron(III) oxide $Fe_2O_3 \cdot xH_2O$ = rust.

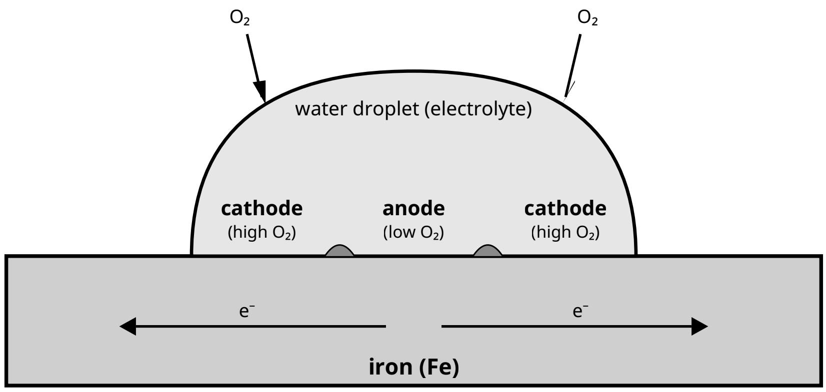

Figure 6. Wet corrosion under a water droplet. Centre (low O_2) = anode (Fe oxidised); edges (high O_2) = cathode (O_2 reduced). Fe^{2+} and OH^- meet to deposit rust.

Electrochemical protection

Method	How it works
Sacrificial anode (Zn/Mg block)	A more reactive metal (more negative E°) is attached and oxidised *instead* of the iron; the iron becomes the cathode. Used on ship hulls/pipelines.
Cathodic protection (impressed current)	An external supply forces the iron to be the cathode so it cannot oxidise.

Galvanising (Zn coat)	Barrier *plus* sacrificial protection — Zn oxidises before Fe even if scratched.
Barrier (paint/oil)	Excludes water and O_2; fails if scratched.
Alloying (stainless steel)	Cr/Ni form a self-repairing protective oxide.

> **WHY SACRIFICIAL PROTECTION WORKS**
> A metal with a **more negative E°** than iron (Zn −0.76, Mg −2.37 vs Fe −0.44) is a stronger reducing agent, so it is oxidised preferentially and protects the iron.

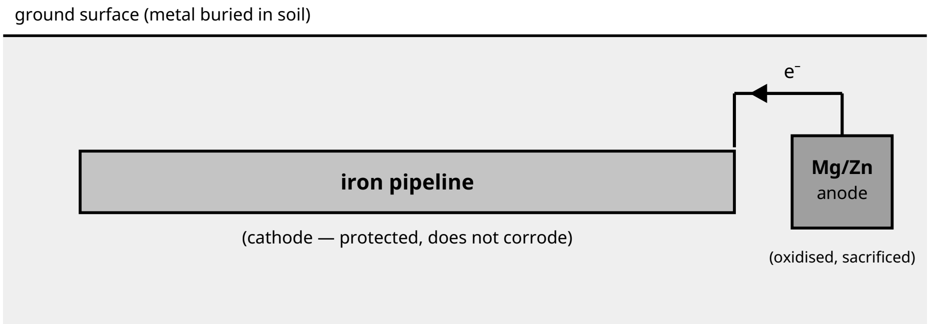

Figure 7. Cathodic (sacrificial) protection. A more reactive metal (Mg or Zn) is connected to the buried iron. The block is the anode and is oxidised ($Mg \rightarrow Mg^{2+} + 2e^-$); its electrons flow into the iron, forcing the iron to be the cathode so it cannot corrode. The sacrificial anode is consumed and replaced periodically.

CHAPTER 10 — ELECTROLYSIS

10.1 Electrolytic Cells

Electrolysis uses electrical energy from an external DC power supply to force a **non-spontaneous** redox reaction to occur ($E°_{cell} < 0$). It is the reverse energy conversion of a galvanic cell: **electrical → chemical** energy.

Feature	Galvanic cell	Electrolytic cell
Energy	chemical → electrical	electrical → chemical
Spontaneity	spontaneous (E°>0)	non-spontaneous (E°<0)
Power supply	none (is the source)	required
Anode sign	negative (−)	**positive (+)**
Cathode sign	positive (+)	**negative (−)**

> **WHAT STAYS THE SAME IN BOTH CELL TYPES**
> Oxidation is **always** at the anode and reduction **always** at the cathode; electrons **always** travel anode → cathode. Only the **polarity (+/−) flips** in an electrolytic cell.

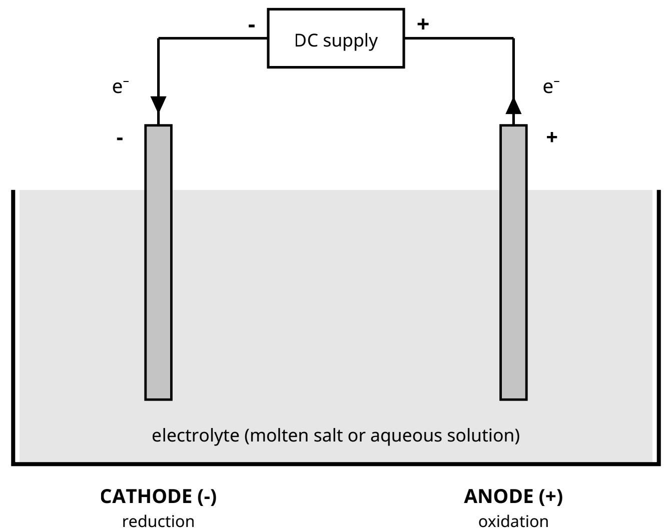

Figure 8. Electrolytic cell. The DC supply pushes electrons onto the cathode (−) and pulls them from the anode (+). Note the polarity is reversed from a galvanic cell, but oxidation is still at the anode and reduction still at the cathode.

At the negative electrode (cathode) — reduction

The cathode is joined to the **negative** terminal, so electrons are pushed onto it. **Cations** (or water) are attracted here and **gain electrons** (reduction). In a molten salt, the metal cation is reduced to the metal.

At the positive electrode (anode) — oxidation

The anode is joined to the **positive** terminal, so electrons are pulled from it. **Anions** (or water) migrate here and **lose electrons** (oxidation).

MOLTEN SODIUM CHLORIDE (NO WATER PRESENT)

$$\text{Cathode }(-)\text{: } Na^+ + e^- \rightarrow Na(l) \quad \text{Anode }(+)\text{: } 2Cl^- \rightarrow Cl_2(g) + 2e^-$$

Only the salt's ions are present, so they are discharged directly: sodium metal forms at the cathode and chlorine gas at the anode.

Competition at the electrodes (aqueous solutions)

In water, **H_2O itself can be reduced or oxidised**, competing with the dissolved ions:

$$\text{Cathode: } 2H_2O + 2e^- \rightarrow H_2(g) + 2OH^- \quad (E° = -0.83\text{ V})$$

$$\text{Anode: } 2H_2O \rightarrow O_2(g) + 4H^+ + 4e^- \quad (\text{from } O_2/H_2O,\ E° = +1.23\text{ V})$$

DECIDING THE PRODUCT AT EACH ELECTRODE

1. List every species that could react there (ions + water).
2. **Cathode:** the species with the **highest E°** is reduced.
3. **Anode:** the species with the **lowest (most negative) E°** is oxidised.
4. Then apply the practical factors below.

Factor	Effect on the product
E° value	The primary predictor.
Concentration	A high ion concentration can cause it to discharge in preference to water (e.g. concentrated $Cl^- \rightarrow Cl_2$ rather than O_2).

Overpotential	Extra voltage needed to release **gases** (especially O_2); this is why Cl_2 often forms instead of O_2.
Electrode type	Inert (Pt, C) is unchanged; an **active** electrode (e.g. Cu) may itself be oxidised at the anode.

EXAMPLE — AQUEOUS $CuSO_4$ WITH INERT ELECTRODES

Cathode: Cu^{2+} (+0.34) vs H_2O (−0.83) → **Cu(s) deposited**

Anode: SO_4^{2-} very hard to oxidise → water oxidised → **O_2(g)** + H^+

Copper plates onto the cathode; O_2 bubbles at the anode; the solution becomes acidic.

Quantitative electrolysis (Faraday's laws)

The amount of substance formed at an electrode is proportional to the electric charge passed.

FORMULAE & METHOD

Formula	Terms
$Q = I \times t$	Q charge (C); I current (A); t time (**seconds**)
$n(e^-) = Q \div F$	$F = 96\,500$ C mol^{-1}
half-equation ratio	relate mol e^- → mol product
$m = n \times M$ / $V = n \times V_m$	convert to mass or gas volume

Method: (1) Q = It → (2) $n(e^-)$ = Q/96 500 → (3) read electron:product ratio → (4) mol product → (5) convert to mass/volume.

WORKED EXAMPLE — SILVER ELECTROPLATING

A current of 2.00 A flows for 30.0 min through $AgNO_3$. Mass of Ag deposited?

$Q = It = 2.00 \times (30.0 \times 60) = 3600$ C

$n(e^-) = 3600 \div 96\,500 = 0.03731$ mol

$Ag^+ + e^- \rightarrow Ag$ (1:1) → n(Ag) = 0.03731 mol

$m = 0.03731 \times 107.9 =$ **4.03 g Ag**

CALCULATION TRAPS

Convert time to **seconds**; use the correct **electron ratio** (Cu^{2+} needs $2e^-$, Al^{3+} needs $3e^-$); cells **in series** pass the **same charge**.

10.2 Industrial Applications of Electrolysis

Electroplating

Coating an object with a thin metal layer. The **object is the cathode**; the **pure plating metal is the anode**; the electrolyte contains ions of the plating metal, so its concentration stays roughly constant as the anode dissolves.

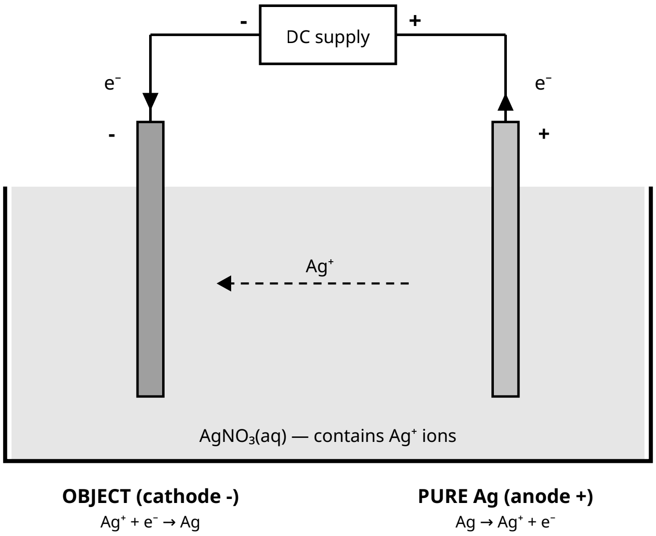

Figure 9. Electroplating with silver. Ag^+ is reduced onto the object (cathode); the pure silver anode dissolves to replenish Ag^+, keeping the solution concentration roughly constant.

Electrorefining of copper

Impure copper is the **anode** (it dissolves); pure copper deposits on the **cathode**. Less-reactive impurities (Ag, Au) fall below as "anode sludge"; more reactive metals stay dissolved.

Production of sodium — the Downs process

Sodium is too reactive to extract by chemical reduction, so **molten NaCl** is electrolysed (with $CaCl_2$ added to lower the melting point).

$$\text{Cathode: } Na^+ + e^- \rightarrow Na(l) \quad \text{Anode: } 2Cl^- \rightarrow Cl_2(g) + 2e^-$$

Production of aluminium — the Hall–Héroult process

Purified aluminium oxide (Al_2O_3) is dissolved in molten **cryolite** (~950 °C) to lower the melting point, then electrolysed.

$$\text{Cathode: } Al^{3+} + 3e^- \rightarrow Al(l) \quad \text{Anode: } 2O^{2-} \rightarrow O_2 + 4e^-$$

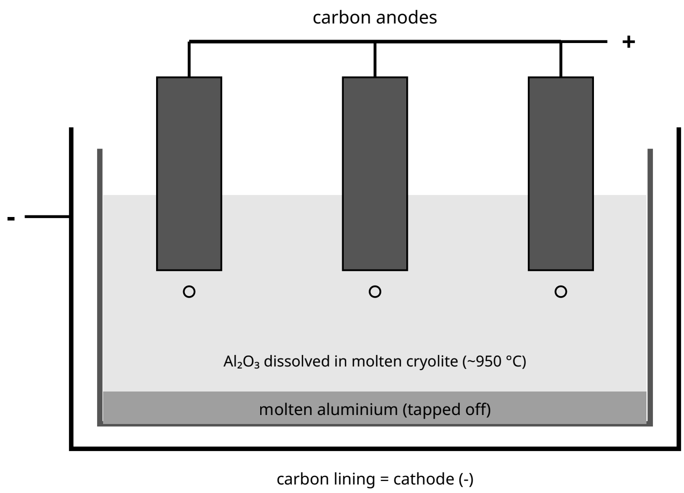

Figure 10. Extraction of aluminium (Hall–Héroult). Cathode (carbon lining): $Al^{3+} + 3e^- \rightarrow Al(l)$, which collects molten at the base and is tapped off. Anode (carbon blocks): $2O^{2-} \rightarrow O_2 + 4e^-$; the hot O_2 burns the carbon anodes ($C + O_2 \rightarrow CO_2$), so they wear away and are replaced.

R1 Exam Traps & Glossary

TOP TRAPS (CHAPTERS 8–9)

- Oxidising agent is **reduced**; reducing agent is **oxidised**.
- Never change E° when multiplying a half-equation (E° is intensive).
- Salt bridge maintains **neutrality/completes the circuit** — it does not carry electrons.
- Always **check charge balance** in half-equations.
- A positive $E°_{cell}$ means **feasible**, not fast.
- More reactive metal = stronger reducing agent = **more negative** E°.

Glossary

Term	Definition
Redox reaction	Electron-transfer reaction (simultaneous oxidation + reduction).
Oxidation / Reduction	Loss / gain of electrons (ox. number increases / decreases).
Oxidising / Reducing agent	Electron acceptor (itself reduced) / donor (itself oxidised).
Oxidation number	Charge an atom would have if all bonds were ionic.
Half-equation	Shows one process (oxidation or reduction) with its electrons.
Conjugate redox pair	Oxidised and reduced forms linked by e^- transfer.
Reactivity series	Metals ordered by ease of oxidation (reducing strength).
Displacement reaction	A more reactive metal reduces the ion of a less reactive metal.
Anode / Cathode	Electrode of oxidation / reduction.
Salt bridge	Maintains neutrality and completes a galvanic circuit.
E° (standard potential)	Reduction potential of a half-cell vs the SHE at standard conditions.
SHE	Standard Hydrogen Electrode; reference defined as 0.00 V.
Electrochemical series	Half-reactions ordered by E° (oxidising strength).
Primary / Secondary cell	Non-rechargeable / rechargeable galvanic cell.
Fuel cell	Galvanic cell with continuously supplied reactants.
Corrosion	Unwanted oxidation of a metal by its environment.

R2 Standard Reduction Potentials (25 °C)

All half-equations written as reductions. Strongest oxidant at top (most positive E°); strongest reductant at bottom (most negative E°). Always use the exact values on your official exam data sheet.

Ordered strongest oxidant (top) → strongest reductant (bottom)

Reduction half-equation	E° (V)	Note
$F_2(g) + 2e^- \rightarrow 2F^-$	**+2.87**	strongest oxidant
$H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$	**+1.77**	
$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$	**+1.51**	acidified permanganate
$Cl_2(g) + 2e^- \rightarrow 2Cl^-$	**+1.36**	
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$	**+1.33**	acidified dichromate
$O_2(g) + 4H^+ + 4e^- \rightarrow 2H_2O$	**+1.23**	
$Br_2(l) + 2e^- \rightarrow 2Br^-$	**+1.09**	

$Ag^+ + e^- \rightarrow Ag(s)$	**+0.80**	
$Fe^{3+} + e^- \rightarrow Fe^{2+}$	**+0.77**	
$O_2(g) + 2H_2O + 4e^- \rightarrow 4OH^-$	**+0.40**	
$Cu^{2+} + 2e^- \rightarrow Cu(s)$	**+0.34**	
$Sn^{4+} + 2e^- \rightarrow Sn^{2+}$	**+0.15**	
$2H^+ + 2e^- \rightarrow H_2(g)$	**0.00**	**SHE reference**
$Pb^{2+} + 2e^- \rightarrow Pb(s)$	**−0.13**	
$Sn^{2+} + 2e^- \rightarrow Sn(s)$	**−0.14**	
$Ni^{2+} + 2e^- \rightarrow Ni(s)$	**−0.25**	
$Fe^{2+} + 2e^- \rightarrow Fe(s)$	**−0.44**	
$Zn^{2+} + 2e^- \rightarrow Zn(s)$	**−0.76**	used in galvanising
$2H_2O + 2e^- \rightarrow H_2(g) + 2OH^-$	**−0.83**	
$Al^{3+} + 3e^- \rightarrow Al(s)$	**−1.66**	
$Mg^{2+} + 2e^- \rightarrow Mg(s)$	**−2.37**	
$Na^+ + e^- \rightarrow Na(s)$	**−2.71**	
$Ca^{2+} + 2e^- \rightarrow Ca(s)$	**−2.87**	
$K^+ + e^- \rightarrow K(s)$	**−2.93**	
$Li^+ + e^- \rightarrow Li(s)$	**−3.04**	strongest reductant

KEY RELATIONSHIP

$E°_{cell} = E°_{cathode} - E°_{anode}$ (positive $\Longrightarrow$ spontaneous galvanic reaction). $\Delta G° = -nFE°_{cell}$, F = 96 500 C mol^{-1}.

Aligned to Pearson Chemistry WA 12, Chapters 8, 9 & 10. Content written as original study notes.